

Rita Bhowmik

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EDUCATION

University Of Waterloo

Waterloo, ON

Mathematics & Economics, Honours with Co-op - 2A Term

Sep. 2025 – May 2030

- **Coursework in** Calc I & II, Calc III, Computing, Statistics, Physics, Microeconomics, Financial Analysis
- **Community Involvement** Waterloo Tech Week, Research @ C&O Department, FARMSA

SKILLS

Languages: Python, HTML/CSS, JavaScript

Frameworks: React, Node.js, Next.js, Vercel, Github, VSCode

Concepts: Data Analysis, UX & UI Development, APIs, Mathematical Modeling, AI Engineering

PROFESSIONAL EXPERIENCE

Engineering Intern

May 2026 – Present

Mespere Lifesciences

Waterloo, ON

- Worked within a technical, fast-paced team designing real-world sensors. Supported manufacturing engineering of R1 and R2 sensor lines, analyzing data and processes to improve reliability and scalability.
- Collaborated with cross-functional teams (software, business, and data) to research sensors with NeuroOS and V0100 software platforms.
- Conducted R&D on photonic hemodynamic sensors, evaluating performance characteristics and contributing to mathematical design improvements for medical device systems

Medical Marketing Analyst

June. 2024 – Sep. 2025

Complete Eyecare Optometry

Mississauga, ON

- Analyzed historical marketing and patient engagement data to design and execute a strategy that generated 40K+ new impressions and increased outreach by 900% over 8 months.
- Implemented SEO optimization strategies, improving search visibility and driving measurable increases in inbound traffic.
- Built and managed analytics dashboards across 4 platforms to track campaign performance and inform data-driven decisions.

AWARDS / PROJECTS

1st Place - Jane Street Estimathon @ Hack the 6ix | *Fermi estimation, Probability*

July 2026

- Won 1st place in the Jane Street Estimathon at Toronto's Hack the 6ix, demonstrating strength in probability, mathematical modeling, and data-driven estimation under strict time constraints.
- Placed 1st out of 120+ competitors, applying probabilistic reasoning, estimation techniques, and quantitative problem-solving.

IMC Prosperity Python Algorithm | *Python, VSCode, Git, Matplotlib, Numpy*

Nov 2025

- Developed algorithmic trading strategies in Python using NumPy and quantitative modeling techniques to optimize decision-making under uncertainty.
- Simulated market conditions and evaluated strategy performance using data-driven backtesting and visualization.
- Applied optimization models to maximize returns within constrained trading environments.